

BEFORE ENGINE START**Familiarise with HANDBOOK first!**

Preflight Inspection COMPLETED

According to Handbook, Cpt. 4.3

Rudder Pedal Position SET

Seat belts TIGHTEN

Flight Controls CHECKED

Canopy LOCKED

Parking Break SET

Apply pressure and set, then 'pump'.

For usage of external Power,
see notes in Handbook Cpt. 4.4.2**ENGINE START**

Circuit Breakers CHECKED

Backup Battery ON

GTN650Xi, MFD, EFIS BOOTING

MFD

Wait for boot sequence.

Check FUEL QTY!

Conf. amber "BACKUP BATT" msg.

Conf. CO value 255 ppm, then decr.

Master Switch ON

MFD

Conf. amber "BACKUP BATT" msg.

CAS message OFF

Note down Total Hours from MFD!

Backup Battery OFF

GEN A/B ON

Voltage M MIN. 12 V

(Batt/Alt-C Bus)

Fuel Selector MORE EMPTY TANK

Propeller Control FINE

Fully forward

Throttle CLOSED

Strobe Light ON

LANE A / LANE B ON

Main Fuel Pump ON

– Continues next column!

CONTINUE: ENGINE START

Start Power ON (HOLD!)

LANE A/B Lamps:

| wait till lamps turn OFF

MFD Engine Indication CHECKED

→ active and correct

→ Fuel Pressure in Limits

Throttle Lever:Coolant/Oil Tmp. $<10^{\circ}\text{C}$ $\geq 55\%$ Coolant/Oil Tmp. $>10^{\circ}\text{C}$ ca. 45 %

% as indicated as "T" on MFD.

See HINT and Start Performance Chart
in Handbook Cpt. 4.4.3 on P. 4-8!**ENGINE START:**

Hold "Start Power", THEN press "START"

until Engine fires. Then release

Start-Button and "Start Power".

Throttle 2000 RPM

Oil Pressure CHECKED

LANE A/B Indicator Lamps OFF

Both LANE SWITCHES remain ON!

Only the Indicator Lamps should go out.

Throttle 2500 RPM FOR 10 SEC

To engage/sync generators.

Then back to 2000 RPM

Voltage M / Voltage B 13.4 / 14.0 V

AFTER ENGINE START

Avionics ON

ALT-C ON

Voltage M CHECKED

Increasing, nom. 13.8 V

EFIS ON

EFIS Power Source CHECKED

Verify NOT on internal Battery

Baro PFD + MFD SET

StbyEFIS, GTN are synced. MFD isn't!

BEFORE TAXIING

Nav Light (at night) AS REQUIRED

Landing-/Taxi-Light ON

A/C moving: Lights on!

PFD, stby EFIS, MFD CHECKED

ADHRS stabilised

TAXIING

Keep RPM < 2500, taxiing with walking speed!
Check BRAKES & STEERING! Ground Control
CHECKED

Brakes CHECKED

Indications CHECKED

Attitude/direction: PFD, MFD, stby EFIS

ENGINE RUN UP

Parking Brake SET

Landing-/Taxi-Light AS REQUIRED

A/C standing still: Lights off!

Engine Warm-Up:

| 2500 RPM until oil temp > 50 °C.

Propeller Control FINE

Fully forward

Pump and Fuel Supply Check

| Throttle: 3000 RPM

AUX Pump: ON, pressure in limits

AUX Pump Light: ON

Main Fuel Pump: OFF

Pressure still in limits: CHECKED

Main Fuel Pump: ON

Fuel Tank SIDES SWITCHED

Fuel Pressure CHECKED

Fuel Tank FULLEST TANK

– Continues next column!

CONTINUE: ENGINE RUN UP**LANE Checks**

| Throttle: set to 2500 RPM

LANE A Switch: OFF

Max. vary ± 250 RPM (aft. 15 sec.)

Check Engine Params in Limits!

LANE A Switch: ON (wait 3 sec.)

LANE B Switch: OFF

Max. vary ± 250 RPM (aft. 15 sec.)

Check Engine Params in Limits!

LANE B Switch: ON (wait 3 sec.)

Propeller Check

| Throttle: set to 4000 RPM

Propeller Control: CHECKED

Cycle 3 times. Move lever SLOWLY
backwards, but fast forward.

Throttle IDLE CHECK

Throttle SET 3000 RPM

BEFORE TAKE-OFF

Instruments CHECKED

Pitot-Heat AS REQUIRED

Flight Controls CHECKED

Cockpit Canopy CLOSED & LOCKED

Seat Belts TIGHTENED

AEPS System UNLOCKED

Store pin in middle compartment!

Engine Instruments CHECKED

Autopilot

| Autopilot Switch: ON

Engage, for each test

Check if AP can be overpowered!

Check AP Disconnect by:

Any Trim input, AP-Disc. button.

Autopilot Switch OFF

Autopilot Panel LEDs ALL OFF

Trim T/O POSITION

Flaps AS REQUIRED

Flaps 0 recommended.

Flaps 1 also possible – See AFM!

TAKE-OFF

Landing-/Taxi-Light ON

Brakes RELEASED

Propeller Control FULL FORWARD

When lined-up:

Throttle FULL POWER

Engine Instruments:

| RPM: min. 5700, max. 5800

Manifold Pressure: within LIMITS

Manifold Temp: Below 65 °C

Airspeed Indication COMES ALIVE

TAKE-OFF SPEEDS

Nosewheel Unstick 50 KIAS

Lift-Off 60 KIAS

Initial Climb 70 KIAS

Flaps retract 75 KIAS

Not below 150 ft AGL!

Climb V_Y 78 KIAS

CLIMB**Climbing Speeds** v_y best ROC 78 KIAS v_x best AOC 63 KIAS

Throttle FULL FORWARD

Prop Control

Set Engine Speed

max. 5800 RPM for 5 min

max. continuous 5500 RPM

Instruments T & P IN LIMITS

AUX Pump OFF

Fuel Pressure CHECKED

Landing Light AS REQUIRED

Outside circuit or CTR: OFF

CRUISE**Fuel Selector**

Switch every 25 min, latest.

Check for balanced quantity!

See notes in AFM Cpt. 4.4.13.

Throttle AS REQUIRED

Prop Control MAX. 5500 RPM

Fuel Flow CHECKED

Pitot Heat AS REQUIRED

Changes in PWR:

Increase Power ↑ 1. RPM – 2. MAP

Decrease Power ↓ 1. MAP – 2. RPM

PWR	RPM	MAP
100 %	5500 RPM	40.5 inHg
75 %	5000 RPM	33.0 inHg
59 %	4500 RPM	25.5 inHg

DECENT

Throttle 15...17 INHG

Prop Control MAX. 5000 RPM

Airspeed 90...100 KIAS

Pitot-Heat AS REQUIRED

Do NOT reduce Throttle to IDLE

in cold OAT environments!

BEFORE LANDING

Seat Belt TIGHTENED

Cockpit PREPARED

Autopilot DISENGAGED

Autopilot Switch OFF

Throttle AS REQUIRED

Speed ≤ 83 KIAS

Flaps FLAPS 1

Trim AS REQUIRED

Landing Light ON

AUX Pump ON

Final Approach

Flaps FLAPS 2

Speed 65 KIAS

Trim AS REQUIRED

Propeller Control FINE

Fully forward

Throttle SET

Approx. 7 inHG,

Maint. RPM > 2600

BALKED LANDING / GO-AROUND

Throttle FULL FORWARD

Speed 70 KIAS

Flaps FLAPS 1

Trim ADJUSTED

Climb V_Y 78 KIAS

Follow T/O Procedures

LANDING

Throttle CLOSED

Touch-down MAIN WHEELS

Brakes AS NEEDED

AFTER LANDING

Flaps RETRACTED

Engine Speed SET

Taxi with walking speed!

Landing-/Taxi-Light ON

Pitot Heat OFF

AUX Fuel Pump OFF

SHUTDOWN

Parking Brake SET
 Landing-/Taxi-Light OFF
 Propeller Control FINE
 Engine Speed IDLE
 Instruments ALL IN LIMITS
 ALT-C OFF

Check Volts M $\geq$ 13.4 V!

min. 2 min COOL-DOWN!

Use timer in PFD or MFD!

Note down Total Hours from MFD!

Lane A OFF
 after 2 sec, LAMP: CHECKED ON
 Lane B OFF

Engine will STOP!

Main Fuel Pump OFF
 GEN A/B OFF
 Avionics OFF
 EFIS OFF
 Strobes OFF
 Master OFF
 AEPS Locking Pin INSERTED
 Parking Brake AS REQUIRED
 All switches are OFF CHECKED

Parking and tie-down:

All Lane & Master Switches OFF
 Parking Brake SET

For short time parking only,
 for long time, use chocks!

Canopy CLOSED & LOCKED

AIRCRAFT SECURED!